

Computer Networks (CS3001)

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Course Instructor(s)

Dr. Ghufraan Ahmed, Dr. Aqsa Aslam,
Dr. Farrukh Salim, Dr. Nausheen Shoaib,
Ms Yusra Kaleem, Ms Hira Tunio,
Ms Rafia Shaikh and Mr Muhammad Usman

Sessional-I Exam

Total Time: 60 mins

Total Marks: 30

Total Questions: 03

Semester: SP-2024

Campus: Karachi

Dept: Computer Science

Student Name

Roll No

Section

Student Signature

Vetted by

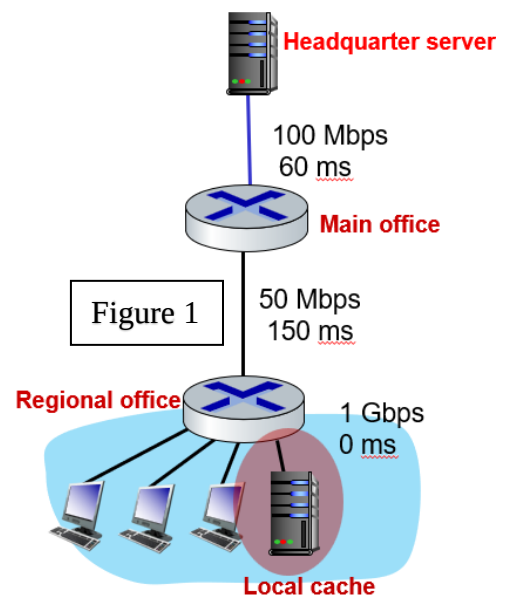
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CLO # 1: Describe and evaluate the protocols, services and functions provided by each layer in the Internet protocol stack

Q1:

[10 marks]

Consider a scenario where a company's headquarters server hosts important software updates and is accessed by employees from their workstations (as shown in Figure 1). The headquarters server is connected to a main office router via a 100 Mbps link with a 60 ms propagation delay. The main office router is connected to a regional office router over a 50 Mbps link with a 150 ms propagation delay. Additionally, the regional router is connected to a host and a cache (if present) via a 1 Gbps link with no propagation delay. All packets in the network are 15,000 bits long.



- How a message would be processed at each layer of the TCP/IP protocol suite in this scenario?
- What is the end-to-end delay from when a packet is transmitted by the headquarters server to when it is received by an employee's workstation? Here, assume there are no caches involved, no queuing delay at the routers, and packet processing delays at routers and nodes are all negligible.
- What is the maximum rate at which the headquarters server can deliver data to a single employee's workstation if we assume no other employees are accessing software updates? Assume that employee

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workstations send requests for files directly to the headquarters server (no caches are used or offline in this case).

- d. What is the average end-to-end delay for the case that 70% of the requests can be satisfied by the local cache servers in the main office?
- e. Outline the steps involved in a conditional GET request and how it reduces network traffic.

CLO #2: Apply network protocol and communication services for client/server and other application layouts

Q2:

[4+3+3=10 Points]

PakTech Solutions is a newly established small business that has acquired the domain name "paktech.com" for their online presence. They have a small network with a few internal resources and users. PakTech Solutions wants to set up DNS to ensure proper domain name resolution and facilitate access to their website, email services, and internal resources.

- a. Illustrate the DNS non-recursive resolution process that occurs when a user at PakTech Solutions attempts to access the company's website by typing "www.paktech.com" into their web browser.
- b. If the IP address for "www.paktech.com" is already stored in the local DNS resolver's cache, explain the procedure by which the user retrieves the IP address in this scenario.
- c. Explain the significance of TTL (Time-to-Live) in DNS. (3-4 lines)

Q3:

[2.5×4=10 Points]

During the evaluation of a new web application, the performance of the website is being assessed by a team of developers. The website contains 20 small images distributed across multiple pages.

- d. Discuss the two advantages and disadvantages of using Persistent and Non-Persistent versions of HTTP for image retrieval. Write 2-3 lines detailing each.
- e. Estimate the total Round-Trip Time (RTT) required to download all images using Persistent and Non-Persistent versions of HTTP.
- f. In the context where a web application features 20 small images spread across various pages, discuss the benefits of utilizing a Peer-to-Peer (P2P) architecture for image distribution. (3-4 lines)
- g. What is the problem of Head-of-Line (HOL) blocking in HTTP/1.1? How does HTTP/2 aim to address this issue?

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